

- Description

This module manages a smart tire tread wear monitoring system, ensuring that:

1. The system continuously measures the tire tread depth (in millimeters).
2. If the tread depth falls below a predefined minimum threshold, an alert is triggered indicating that the tire needs replacement.
3. If the tread depth is above the threshold, the tire is considered safe for use.

This system helps ensure vehicle safety by providing timely alerts when tire tread wear reaches critical levels.

- Specification

module SmartTireTreadWearMonitoring

var currentTreadDepth, minTreadDepth: real

process updateTreadDepth(newDepth: real) returns (currentTreadDepth: real)

functional scenario form:

currentTreadDepth < minTreadDepth && output "ALERT: Tire tread depth (" + currentTreadDepth + " mm) is below safe threshold (" + minTreadDepth + " mm). Tire replacement required.", Tire tread is below safety threshold; replacement alert triggered.

||

currentTreadDepth >= minTreadDepth && output "Tire tread depth (" + currentTreadDepth + " mm) is within safe limits.", Tire tread is acceptable.

end process

end module

- Java Code

```
package code;
public class c106_SmartTireTreadMonitor {
    private double minTreadDepth;
    private double currentTreadDepth;
    public c106_SmartTireTreadMonitor(double minTreadDepth, double initialTreadDepth) {
        this.minTreadDepth = minTreadDepth;
        this.currentTreadDepth = initialTreadDepth;
    }
    public void updateTreadDepth(double newTreadDepth) {
        currentTreadDepth = newTreadDepth;
        System.out.printf("Current Tire Tread Depth: %.2f mm%n", currentTreadDepth);
        if (currentTreadDepth < minTreadDepth) {
            System.out.println("ALERT: Tire tread depth too low! Tire replacement required.");
        } else {
            System.out.println("Tire tread depth is within safe limits.");
        }
    }
}
```

```

    }
}
public double getCurrentTreadDepth() {
    return currentTreadDepth;
}
public static void main(String[] args) {
    c106_SmartTireTreadMonitor monitor = new c106_SmartTireTreadMonitor(3.0, 4.5);
    // testcase-VT:
    //     monitor.updateTreadDepth(4.0);
    //     monitor.updateTreadDepth(2.5);
    //     monitor.updateTreadDepth(3.5);

    // testcase-FT:
    //     monitor.updateTreadDepth(-4.864346129627444E307);
    //     monitor.updateTreadDepth(0.0);
    //     monitor.updateTreadDepth(-1.7976931348623131E308);

    // testcase-Z3:
    //     monitor.updateTreadDepth(4.0);
    //     monitor.updateTreadDepth(2.5);
    //     monitor.updateTreadDepth(3.5);

    // testcase-UVT:
    monitor.updateTreadDepth(4.0);
    monitor.updateTreadDepth(2.5);
    monitor.updateTreadDepth(3.5);
}
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c106_SmartTireTreadMonitor.java	100.0 %	2	0	2
▼ c106_SmartTireTreadMonitor	100.0 %	2	0	2
● updateTreadDepth(double)	100.0 %	2	0	2
● c106_SmartTireTreadMonitor(dc		0	0	0
● main(String[])		0	0	0

3. Vibration testing instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	54	21,889	21,943
▼ src	0.2 %	54	21,889	21,943
▼ code	0.2 %	54	21,889	21,943
▼ c106_SmartTireTreadMonitor.java	94.7 %	54	3	57
▼ c106_SmartTireTreadMonitor	94.7 %	54	3	57
● updateTreadDepth(double)	100.0 %	29	0	29
● c106_SmartTireTreadMonitor(dc	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.1 %	1	841	842
▼ src	0.1 %	1	841	842
▼ code	0.1 %	1	841	842
▼ c106_SmartTireTreadMonitor.java	50.0 %	1	1	2
▼ c106_SmartTireTreadMonitor	50.0 %	1	1	2
● updateTreadDepth(double)	50.0 %	1	1	2
● c106_SmartTireTreadMonitor(dc		0	0	0
● main(String[])		0	0	0

5. Fuzz testing instruction coverage: 89.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	51	21,892	21,943
▼ src	0.2 %	51	21,892	21,943
▼ code	0.2 %	51	21,892	21,943
▼ c106_SmartTireTreadMonitor.java	89.5 %	51	6	57
▼ c106_SmartTireTreadMonitor	89.5 %	51	6	57
● updateTreadDepth(double)	89.7 %	26	3	29
● c106_SmartTireTreadMonitor(dc	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c106_SmartTireTreadMonitor.java	100.0 %	2	0	2
▼ c106_SmartTireTreadMonitor	100.0 %	2	0	2
● updateTreadDepth(double)	100.0 %	2	0	2
● c106_SmartTireTreadMonitor(dc		0	0	0
● main(String[])		0	0	0

7. Z3 instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	54	21,889	21,943
▼ src	0.2 %	54	21,889	21,943
▼ code	0.2 %	54	21,889	21,943
▼ c106_SmartTireTreadMonitor.java	94.7 %	54	3	57
▼ c106_SmartTireTreadMonitor	94.7 %	54	3	57
● updateTreadDepth(double)	100.0 %	29	0	29
● c106_SmartTireTreadMonitor(dc	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c106_SmartTireTreadMonitor.java	100.0 %	2	0	2
▼ c106_SmartTireTreadMonitor	100.0 %	2	0	2
● updateTreadDepth(double)	100.0 %	2	0	2
● c106_SmartTireTreadMonitor(dc		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 94.7%

Element		Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼  VibrationTestingTool		 0.2 %	54	21,889	21,943
▼  src		 0.2 %	54	21,889	21,943
▼  code		 0.2 %	54	21,889	21,943
▼  c106_SmartTireTreadMonitor.java		 94.7 %	54	3	57
▼  c106_SmartTireTreadMonitor		 94.7 %	54	3	57
 updateTreadDepth(double)		 100.0 %	29	0	29
 c106_SmartTireTreadMonitor(dc		 100.0 %	9	0	9
 main(String[])		 100.0 %	16	0	16